

SUWON, Korea, Republic of

WMO: 471190

Lat: 37.2722N

Lon: 126.9853E

Elev: 124

StdP: 14.63

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.7	15.2	-6.4	3.9	17.7	-2.6	4.8	20.3	14.4	26.6	13.0	27.3	3.6	320	0.387

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.0	90.9	77.3	88.5	76.2	86.2	74.7	79.7	86.2	78.6	84.7	77.5	83.3	6.3	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.1	146.8	82.4	76.9	140.9	81.9	75.8	135.7	81.1	43.5	86.1	42.3	84.9	41.2	83.4	90.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.6	11.7	10.1	7.3	94.1	4.2	2.6	4.3	96.0	1.8	97.6	-0.6	99.0	-3.6	101.0
			5.5	81.5	4.0	2.6	2.6	83.3	0.3	84.8	-2.0	86.3	-4.9	88.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.2	28.4	32.9	42.5	54.1	64.3	72.7	78.4	79.6	71.1	59.2	45.5	32.5
	DBStd	18.73	7.21	6.83	6.48	6.38	4.76	3.68	3.91	4.41	4.65	6.29	7.63	7.42
	HDD50	2154	670	480	246	31	0	0	0	0	0	10	176	541
	HDD65	4928	1135	900	698	331	70	2	0	0	7	193	586	1006
	CDD50	4061	0	1	13	153	444	682	879	919	633	297	40	0
	CDD65	1361	0	0	0	3	50	234	414	454	191	15	0	0
	CDH74	10509	0	0	0	34	336	1356	3420	4298	1016	48	1	0
	CDH80	3063	0	0	0	3	43	282	1040	1517	177	1	0	0
Wind	WSAvg	4.0	3.6	4.0	4.7	4.7	4.3	4.1	4.2	4.2	3.7	3.3	3.5	3.6
Precipitation	PrecAvg	51.0	1.0	1.2	1.9	3.6	3.4	4.7	13.2	11.0	5.8	2.2	2.1	0.9
	PrecMax	80.5	3.5	4.6	5.0	15.6	9.6	16.3	22.4	33.5	23.0	10.4	5.9	2.1
	PrecMin	33.5	0.1	0.0	0.2	0.3	0.5	0.2	5.7	2.1	0.1	0.0	0.4	0.0
	PrecStd	10.7	0.9	1.0	1.1	3.4	2.1	4.1	5.5	6.1	4.7	2.4	1.4	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.5	57.0	66.8	78.0	84.4	87.9	93.8	95.0	87.2	78.3	68.2	54.3
		MCWB	43.5	49.1	53.1	59.9	64.5	69.5	76.9	78.9	73.5	64.4	58.1	46.6
	2%	DB	44.8	51.0	61.2	73.4	80.6	85.4	90.3	91.9	84.3	75.0	63.9	49.9
		MCWB	38.3	42.9	49.8	57.9	63.1	69.3	77.0	77.8	71.3	62.2	55.2	43.8
	5%	DB	41.8	47.3	57.4	69.8	77.8	83.4	87.9	89.8	81.9	72.1	60.4	46.6
		MCWB	36.2	39.7	46.8	54.7	61.8	68.7	76.6	77.3	69.6	61.2	52.8	41.3
	10%	DB	39.1	43.7	53.8	66.2	74.9	81.2	85.5	87.5	79.7	69.4	57.4	43.8
		MCWB	34.4	37.4	44.9	53.5	60.4	67.8	75.5	76.4	68.6	59.6	51.1	38.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.2	51.2	56.3	64.9	69.0	75.2	81.0	81.5	77.7	69.4	61.5	48.7
		MCDB	48.8	54.5	62.6	73.8	78.8	80.5	86.7	88.8	82.3	72.4	65.0	51.4
	2%	WB	40.2	43.9	52.3	60.8	66.4	73.3	79.5	80.1	75.3	65.7	57.4	44.8
		MCDB	43.6	49.1	58.6	70.0	74.2	78.9	85.4	87.1	80.1	71.0	61.5	48.7
	5%	WB	37.3	40.9	48.6	57.8	64.8	71.9	78.3	79.1	73.5	63.6	54.5	41.8
		MCDB	40.7	45.8	54.7	65.8	72.7	78.2	84.1	85.6	78.1	69.3	58.7	46.0
	10%	WB	34.6	38.1	45.8	55.3	63.1	70.5	77.3	78.2	71.5	61.5	51.8	39.3
		MCDB	38.4	43.0	52.5	63.2	71.4	77.3	82.6	84.4	76.7	67.3	56.3	43.5
Mean Daily Temperature Range	5% DB	MDBR	14.7	16.0	17.2	18.9	18.1	15.0	11.3	12.0	14.6	17.4	15.7	14.5
		MCDBR	16.8	20.5	22.9	24.4	22.9	18.8	14.7	14.7	17.2	19.5	19.0	17.3
		MCWBR	11.4	12.8	12.4	10.9	8.5	6.4	5.2	5.1	6.4	8.4	11.4	12.1
	5% WB	MCDBR	15.0	17.8	19.2	19.7	17.7	13.6	11.6	12.6	12.5	15.6	15.6	16.3
		MCWBR	11.8	13.0	12.6	11.2	8.9	6.2	5.2	5.4	6.3	8.9	11.5	12.7
Clear-Sky Solar Irradiance	taub	0.440	0.490	0.572	0.613	0.626	0.698	0.662	0.582	0.510	0.499	0.482	0.437	
	taud	1.948	1.859	1.715	1.675	1.697	1.619	1.740	1.930	2.054	1.957	1.944	1.975	
	Ebn at Noon	220	226	220	221	221	206	212	226	236	221	205	211	
	Edh at Noon	44	55	69	76	76	82	72	58	49	49	44	41	
All-Sky Solar Radiation	RadAvg	789	1035	1358	1581	1774	1651	1306	1385	1308	1140	778	692	
	RadStd	49	84	136	114	164	167	197	198	142	82	79	48	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	+0.88	+0.95	N/A	N/A	+126	N/A
(47)	Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+169	+90

Nomenclature: See separate page